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Competition, coalitions and canine size in primates

Explanations of the evolution of canine tooth size sexual dimorphism in primates implicitly assume that male canine size is the primary target of natural or sexual selection, even though there is substantial unexplained variation in female canine size. Here we demonstrate that the canines of both males and females are relatively larger in species where agonistic interactions are more likely to escalate. Hence, the canine teeth of both sexes are influenced by selection for the development of weaponry, though the nature of the selection (natural or sexual) favoring development of the canines as weapons depends on the objects of competition (mates *vs.* food, for example). However, we also suggest that where the outcome of fights is typically determined between coalitions of individuals, selection for weaponry is reduced, even though agonistic competition can be very intense. Coalitionary aggression between males occurs in only a handful of anthropoid primates, but it is common among female anthropoids, and helps explain, in part, the extreme sexual dimorphism in the canine teeth of a number of catarrhine species. This is the first demonstration that cooperation among animals may affect their morphology. The lack of sexual dimorphism in canine size among prosimian primates is also consistent with the model. This analysis demonstrates that a better understanding of the interspecific variation in sexual dimorphism among primates is gained by considering behavioral variation in both sexes.

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Introduction

Comparative studies generally corroborate the hypothesis that sexual selection—specifically male-male competition for access to mates—is at least partly responsible for the evolution of sexual dimorphism in body size and weaponry in mammals (Bartholomew, 1970; Crook, 1972; Ralls 1976; Leutenegger & Kelly, 1977; Clutton-Brock & Harvey, 1978; Harvey *et al.*, 1978*b*; Alexander *et al.*, 1979; Kay *et al.*, 1988; Greenfield, 1992*c*; Plavcan & van Schaik, 1992). A number of studies, however, have pointed out that most sexual selection models are incomplete because they ignore variation in females, even though the expression of dimorphism is potentially affected by changes in either male or female traits (Brooks, 1991; Ralls, 1976, 1977; Richard, 1992; Greenfield, 1992*a*; Shea, 1986).

Many primates show strong sexual dimorphism in the canine teeth, with males possessing larger canines than females. Canine tooth size dimorphism in primates has long been thought to be a product of sexual selection (Darwin, 1871; Crook, 1972; Leutenegger & Kelly, 1977; Harvey *et al.*, 1978*b*; Leutenegger, 1982; Kay *et al.*, 1988; Greenfield & Washburn, 1991; Plavcan & van Schaik, 1992). Since males are theoretically limited in reproductive capacity by the number of females they can inseminate, the reproductive pay-offs should be high for those males who can exclude others from access to females. Therefore selection should favor the development of any feature, such as large canine teeth that aids individuals in winning contests. That male primates engage in fights (with occasionally lethal consequences—Kunkun, 1986; Palombit, 1993) and often show evidence of wounding supports the notion that canine teeth are used as weapons in intrasexual combat.

Past comparative analyses attempted to test this hypothesis by using broad classifications of mating system (monogamy; polyandry; polygyny; polygynandry) as a surrogate measure of the strength of intermale competition, and hence sexual selection. However, such analyses only

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weakly corroborated the sexual selection hypothesis. The hypothesis is supported in that monogamous species are monomorphic in canine tooth size, while non-monogamous species show various degrees of canine tooth size dimorphism (Leutenegger & Kelly, 1977; Harvey *et al.*, 1978*b*; Leutenegger, 1982; Kay *et al.*, 1988; Plavcan & van Schaik, 1992). Yet among the non-monogamous species, mating system accounts for only part of the variation in canine dimorphism, with some species showing little or no dimorphism and others intense dimorphism (Phillips-Conroy & Jolly, 1981; Rowell & Chism, 1986; Plavcan & van Schaik, 1992). Prosimian primates provide perhaps the best example of the lack of congruence between mating system and dimorphism, showing a variety of mating systems, even though nearly all are monomorphic in body size (Kappeler, 1990*a*, 1992) and canine size.

Perhaps because of the poor relation between mating system and sexual dimorphism, a number of studies have debated the role of sexual selection in the evolution of sexual dimorphism (in not only canine tooth size, but also body size), advocating important roles for other factors such as diet (Milton, 1985; Greenfield, 1992*a*,*b*), predation defense (Harvey *et al.*, 1978*b*), body size (Leutenegger & Cheverud, 1982, 1985), and phylogenetic inertia (Cheverud *et al.*, 1986; Lucas *et al.*, 1986; Greenfield & Washburn, 1991). While these factors are not mutually exclusive, and in fact there is evidence that each plays at least some role in the evolution of canine dimorphism, more recent analyses demonstrated that more direct estimates of the strength of male–male competition, rather than broad estimates of mating system, explain a large proportion of the variation in canine dimorphism and male canine size in anthropoid primates (Kay *et al.*, 1988; Plavcan & van Schaik, 1992; Plavcan, 1993), offering support for the sexual selection hypothesis. One problem with using mating system to estimate the strength of sexual selection is that mating system is usually inferred from group composition, yet group composition can be misleading for this purpose. For example, group composition cannot account for differences in the observed tolerance of males toward one another, while some multimale *Eulemur* groups in fact may not be characterized by a multimale mating system (van Schaik & Kappeler, 1993). These latter studies emphasize that the strength of selection for the development of weaponry should be proportional to some measures of the frequency and intensity of contest competition among individuals, which are not directly related to broad measures of mating system. It is important to note that these studies do not exclude other factors such as predation or phylogenetic inertia from influencing the evolution of dimorphism in primates.

Almost all prior comparative studies of primate canine dimorphism either implicitly or explicitly focus on the male as the target of sexual selection, with females considered as little more than a baseline of comparison. This is surprising, because female canine size is nearly as variable among species as that of males (Plavcan, 1990), implying that variation in dimorphism is at least partly a function of variation in female canine size. That females sometimes fight just as intensely as males, with lethal consequences having been documented (Gust & Gordan, 1991) suggests that canines might also be useful to females in agonistic competition. This suggests that a better understanding of the evolution of canine dimorphism might be gained by studying the factors that influence canine size in both sexes, rather than focusing exclusively on males.

Available studies of the causes of interspecific variation in female canine size offer contradictory and inconclusive results. Harvey *et al.* (1978*a*) suggest that female canine size may be related to resource competition with males, since females living in multi-male groups have larger canines than those living in single-male groups. However, Smith (1981) produces exactly the opposite result as Harvey *et al.* (1978*a*) by using a measure of canine crown height

rather than basal dimensions of the tooth. Lucas *et al.* (1986) suggest that female canine size should be related to social structure and behavior, but provide no specific model or comparative analysis. Greenfield (1992*a*,*b*), proposes that variation in female competition plays little role in the evolution of female canine size, and suggests that much interspecific variation in female canine size represents a correlated response (Lande, 1980) to variation in male canine size, but that female canines are also selected to function like the incisors for gripping, pulling and cutting food rather than as weapons. To date, no study has attempted to test the hypothesis that, as for males (Greenfield, 1992*b*,*c*; Plavcan, 1993), female canine teeth are used as weapons in agonistic intrasexual competition. Many female primates compete just as intensely and frequently as males, as shown by rates of aggression (e.g. Smuts, 1987). If large canine teeth benefit males in winning contests, they should also benefit females. Therefore, if estimates of agonistic male–male competition are associated with variation in male canine tooth size, then estimates of agonistic female–female competition should be associated with variation in female canine tooth size.

Additionally, Plavcan & van Schaik (1992) previously hypothesized that when agonistic competition typically occurs between alliances (or coalitions) rather than between individuals acting alone, selection for the development of weaponry should be relaxed. While coalitionary competition is relatively rare in male primates, many female primates compete regularly in alliances, offering an ideal opportunity to test this hypothesis.

In this paper, we reconsider the evolution of canine dimorphism in primates, testing the hypothesis that canine tooth size in both sexes is associated with estimates of the frequency, intensity and context (coalitionary/non-coalitionary) of agonistic intrasexual competition, especially when adjusted for body size and phylogeny. We also consider the alternative hypotheses that canine size in either sex is related to substrate (as a surrogate measure of predation pressure) and diet.

Competition

Before presenting our analysis, we briefly review the model of competition frequency and intensity as presented in Plavcan & van Schaik (1992), generalizing it to include females, and describe the model for coalitionary competition.

If large canine teeth are selected for their utility in intrasexual combat, then variation in canine tooth size should be correlated with variation in the frequency and intensity of intrasexual contest competition (Clutton-Brock, 1985) regardless of sex. Unfortunately, comparative data on the actual rates and intensity of such competition are rare. However, the intensity and potential frequency of intrasexual contest competition can be estimated using two assumptions: first that contest intensity is inversely proportional to the observed tolerance of individuals toward one another, and second that the potential frequency of contests is proportional to the proximity of individuals to each other (Plavcan & van Schaik, 1992). Plavcan & van Schaik (1992) estimated these parameters for male–male contest competition (roughly proportional to that of females). We generalize these estimates of intrasexual contest competition to males and females.

We dichotomize contest competition intensity into high and low classes. *High-intensity* contest competition is recognized when adults are either intolerant of one another, or when stable dominance hierarchies are maintained through agonistic interactions. In contrast, *low-intensity* contest competition is recognized when adults are relatively tolerant of one another with normally peaceful relations, or when escalated aggression is rare and dominance hierarchies, if present at all, are reported as difficult to detect. It is important to note that these definitions

are not meant to imply that low-intensity species do not compete at all. Rather, the definitions are intended to draw a broad distinction between observed levels of competition. It should also be noted that, practically, these distinctions roughly parallel the distinction between "contest" and "scramble" competition currently recognized in primate behavioral theory (van Schaik & van Noordwijk, 1988; van Schaik, 1989).

Likewise, we dichotomize potential contest frequency into high and low classes. A high potential frequency of contest competition (*high-frequency*) is recognized whenever two or more adults of the same sex regularly occur together in breeding groups. *Low-frequency* contest competition is recognized whenever adults are either solitary, or only one adult of a single sex typically occurs in a breeding group. In the low-frequency class, contests should be generally limited to at most daily contacts with neighboring or transient individuals. On the other hand, individuals in the high-frequency class are exposed to competitors much more frequently. Species in which multimale influxes are reported are tentatively classified as low-frequency since such influxes usually only occur during a brief mating season, and do not necessarily occur for every breeding group (Henzi & Lawes, 1988). Short of actual comparative data on the frequency of contest competition, this dichotomy represents the only practical definition of frequency that can be used in a broad comparative analysis.

We hypothesize that when escalated competition occurs between alliances (or coalitions), rather than between individuals, the outcome of conflicts should be determined by both the size of the alliance and some weighted sum of individual weaponry. It follows that the value of individual weaponry in determining the outcome of fights should be relatively less than when individuals compete alone, and that selection for the development of weaponry should be weaker (Plavcan & van Schaik, 1992). For the purposes of testing this hypothesis, we only consider the effect of coalitions when escalated competition is *primarily* coalitionary. For example, if males are reported to aggressively defend a territory as a group (indicating coalitionary competition), but individuals sort themselves into a dominance hierarchy on the basis of individual fighting, we do *not* classify the species as coalitionary because selection related to competition for position within the dominance hierarchy should favor the development of large canines. Following this, it is critical to understand that the effect of coalitions should not be viewed as selection "for" smaller canines (or for canine reduction), but rather as a circumstantial mechanism which reduces the relative value of weaponry in winning contests. Thus, coalitionary competition should be associated with a reduction in selective pressure to increase canine size in association with intense fighting, resulting in either a lack of increase in canine size, or an actual reduction in canine size if there are concomitant pressures (selective or non-selective) to reduce canine size (see discussion).

It is important to note that, for the purpose of hypothesis testing, the relative canine size of coalitionary species can only be compared to that of high-frequency, high-intensity species. Since coalitionary competition is hypothesized to moderate the effect of escalated competition on the development of weaponry, we can only predict that such species should have smaller canines than other high-intensity species. We cannot make an *a priori* prediction that canines in these species should be larger or smaller than those of low-intensity species. Likewise, if the effect of frequency is independent of intensity (Plavcan & van Schaik, 1992), then coalitionary species cannot be compared to low-frequency species, regardless of the intensity classification.

In practice, we classified each sex of each species as showing *coalitionary* or *non-coalitionary* competition, depending on whether competition typically occurs between alliances of related individuals or not. For example, in macaques, *Cercopithecus aethiops*, and baboons, a female's

position in the dominance hierarchy is determined to a large degree by her mother's rank (Gouzoules & Gouzoules, 1987; Walters & Seyfarth, 1987). Fights are common, but the outcome is often determined by the (potential) enlistment of support by alliance partners. Among males coalitionary competition is possibly important only in "male-bonded" species, specifically *Pan troglodytes*, *Ateles*, and *Brachyteles*. It is important to note that these species are classified as competing in coalitions based on intergroup relations, and that within-group relations would imply low-intensity, high-frequency competition. For the purposes of hypothesis testing, though, we must classify these species as showing high-intensity, high-frequency competition. In one other species, *Colobus badius* (Smuts, 1987), males do not usually transfer out of their natal troops, suggesting a condition similar to these other species. However, males are also reported to form an agonistic dominance hierarchy (Struhsaker, 1975; Isbell, 1984). To render the analysis conservative, we tentatively classified this species as coalitionary. Finally, occasional coalitions are formed by male baboons, but these alliances are transient, while top-dominance is achieved individually (Noë, 1990). Since most escalated fights are between individual male baboons, the effect of coalitionary competition is ignored.

It is important to recognize that all of these definitions reflect *proximate* measures of behaviors that should reflect selection (either natural or sexual) for the development of weaponry in either sex. It is well known that males usually compete for mates, while females usually compete for resources (Trivers, 1972; Wrangham, 1980). Thus the reproductive consequences of winning and losing fights, which ultimately should determine the strength of selection for the development of weaponry, are probably greater in males than in females (as noted by Greenfield, 1992*a*). Therefore, while our model predicts that canine size in both sexes should be associated with agonistic intrasexual competition, it does not predict that competition in either sex should produce quantitatively identical effects on the development of weaponry. Following this, our analysis emphasizes first the effects of competition within each sex separately, then the effects of competition within each sex on canine tooth size dimorphism.

Finally, we emphasize that our classifications are devised to reflect broad distinctions among species in either male–male or female–female agonistic competition. In fact, all primates compete, and the nature of competition varies widely between the sexes and from species to species. Our classifications have been devised in such a way as to test the basic hypothesis that relative canine tooth size in both males and females reflects selection for the development of weaponry, and should not be interpreted as indicating that males or females of any one species do not compete. Thus, for example, we recognize that while the dispersion of female orang-utans may reflect competition between individuals for resources, relations between female orang-utans tend, nevertheless, to be relatively peaceful.

Materials and methods

Metric data

Dental metric data were gathered for 94 species of non-human primates housed in museums throughout Europe and North America (Plavcan, 1990; Kappeler, 1992). Canine tooth crown height (the distance from the cementum-enamel junction to the apex of the canine) was measured for the maxillary canine teeth. For interspecific comparisons this aspect of the tooth (as opposed to measurements of the base of the canine) is most variable in both males and females, and is thought to be the most important aspect for the tooth's function as a weapon (Plavcan, 1990, 1993; Greenfield & Washburn, 1991). Detailed descriptions of the

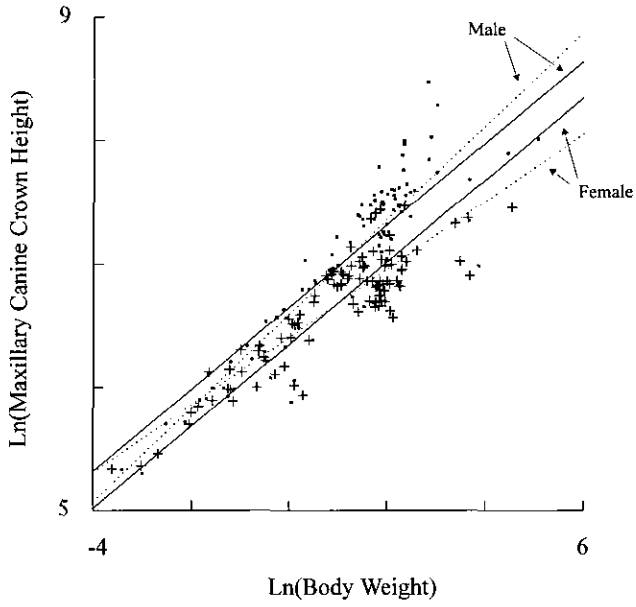


Figure 1. Bivariate plot of ln-transformed male and female maxillary canine crown height *vs.* ln-transformed male and female body weight. Dots indicate males, crosses indicate females. Lines indicate least-squares regressions for males and females separately (dotted lines), and isometric lines passed separately through the male and female data (solid lines).

measurement techniques and samples are provided elsewhere (Plavcan, 1990, 1993; Kappeler, 1992). Body weights for anthropoids were taken from Plavcan & van Schaik (1992), and for prosimians from Kappeler (1992).

Relative canine size was estimated in males and females in two ways (Figure 1). First, we calculated residuals from least-squares regressions of ln-transformed canine size *vs.* ln-transformed body weight for each sex separately (residuals from reduced major axis regressions produced similar results). Next, we calculated residuals from an isometric line (slope=0·333, since canine size is a linear measure, while body weight is volumetric) passed through a data set of both males and females (effectively treating the sexes as separate populations) (Table 1). We preferred this latter method for two reasons. First, male canine size does not scale significantly differently from isometry, particularly when scaling relations are considered within families and subfamilies (Plavcan, 1990, 1993). Second, interactions between behavioral variables, body size, and phylogeny clearly skew allometric relations, especially for the pooled-sex comparisons. We therefore feel that, for the purposes of this study, relative canine size is best measured against the theoretical line representing geometric scaling. Following this, we present values from isometric comparisons (Table 1). Nevertheless, we performed all analyses using both residuals from empirical regressions (least squares or reduced major axis) and residuals from isometric lines. Results from both methods produced the same conclusions.

Behavioral and ecological classifications

Species were classified by intensity, potential frequency and context (coalitionary/non-coalitionary) using literature reports of investigations of wild populations whenever possible

(Table 1). Initial classification of anthropoids was provided by J.M.P. and C.P.vS. then re-evaluated the classifications without prior knowledge of the canine data. Ambiguous cases were clarified by directly contacting field observers. The classification of prosimians was provided by P.M.K., and was reviewed by the other two authors for consistency with the anthropoid classification before the canine data were analysed. Because literature reports providing the information necessary to classify competition were not available for one or both sexes for a number of species, sample sizes vary among the analyses that follow in the results section.

Species' behaviors were classified on the basis of both within- and between-group behavior. This produced some surprising classifications. For example we classified *Aotus* females as showing high-intensity competition on the basis of reports that females aggressively defend territories (Wright, 1985, pers. comm.), even though the species is clearly monogamous and intragroup relations are peaceful. Where observations of competition types conflicted, species were classified by that behavior hypothesized to reflect selection for enhanced weaponry. For example, if the males or females of a species show intragroup competition that is typically high-intensity but non-coalitionary, and intergroup competition that is typically coalitionary, we favored a non-coalitionary classification because the high-intensity, non-coalitionary competition should produce selection for the development of weaponry. In all cases, we rendered the classifications as conservative as possible.

It has long been hypothesized that large canines are selected for defense against predators in terrestrial species (Harvey *et al.*, 1978*a,b*; Plavcan & van Schaik, 1992). If males are normally responsible for defending groups from predators, then selection should favor the development of canine dimorphism since long sharp canine teeth are potentially useful as weapons against predators. Several studies have noted that terrestrial and savanna-dwelling species tend to show more dimorphism (in either body size or canine size) than arboreal species (Harvey *et al.*, 1978*a,b*; Phillips-Conroy & Jolly, 1981; Gautier-Hion & Gautier, 1985; Plavcan & van Schaik, 1992). Assuming that terrestrial, and especially savanna, species need to defend themselves against large-bodied predators more than arboreal species (which can flee in the trees), this is taken as support for this hypothesis. Critical to this hypothesis is the assumption that large canines are not particularly useful in defending against raptors, which commonly prey on arboreal primates (Struhsaker & Leaky, 1990). Species were classified by substrate (arboreal, arboreal/terrestrial, savanna) following Plavcan & van Schaik (1992), Charles-Dominique (1977), and Tattersall (1982).

Dietary data for anthropoids were taken from Plavcan (1990), and for prosimians from Bearder (1987). Species were classified as folivorous, folivorous/frugivorous, frugivorous/folivorous, frugivorous, and frugivorous/insectivorous on the basis of relative proportions of each component in the diet (Plavcan & van Schaik, 1992). Two hypotheses have been put forward predicting a *general* relation between diet and canine size and/or dimorphism. First Milton (1985) argues that folivores are energetically constrained from intrasexual competition, and therefore should show lower degrees of canine dimorphism than non-folivorous species. Second, Greenfield (1992*b*) hypothesizes that canines in both males and females are under continuous selection for incorporation into the incisal "functional field", resulting in canine reduction in the absence of selection to maintain the canines as weapons. This latter hypothesis makes no explicit prediction about a relation between canine size and diet. However, Hylander (1975) observes that folivores have smaller incisors than frugivores, probably because incisors are more important to frugivores in food preparation. Following this, we might predict that frugivores should have smaller canines than folivores if the Greenfield hypotheses were true.

Table 1 Behavioral classifications and residual maxillary canine crown height

Species	Males			Females			Residuals[a]		References
	C[b]	I	F	C	I	F	Males	Females	
					Catarrhines				
Cercocebus albigena	No	High	High	Yes	High	High	0·153	−0·367	Chalmers (1968*a*,*b*); Chalmers & Rowell (1971); Waser (1984)
Cercocebus torquatus	No	High	High	Yes	High	High	0·487	−0·208	Bernstein (1976)
Cercopithecus aethiops	No	High	High	Yes	High	High	0·421	−0·013	Henzi (1985); Whitten (1983)
Cercopithecus ascanius	No	High	Low	No	High	High	0·408	0·085	Cords (1987, 1988); Jones & Bush (1988); Rowell (1988)
Cercopithecus cephus	No	High	Low	—	—	—	0·428	−0·027	Gautier-Hion & Gautier (1985)
Cercopithecus lhoesti	No	High	Low	—	—	—	0·383	−0·029	Cords (1988)
Cercopithecus mitis	No	High	Low	No	High	High	0·396	−0·096	Rowell (1972)
Cercopithecus mona	No	High	Low	No	High	High	0·489	0·035	Kirkevold & Crocket (1987)
Cercopithecus neglectus	No	High	Low	—	—	—	0·405	0·062	Cords (1988); Gautier-Hion (1978); Brennan (1985)
Cercopithecus nicitans	No	High	Low	—	—	—	0·371	0·044	Gautier-Hion & Gautier (1985)
Cercopithecus pogonias	No	High	Low	—	—	—	0·360	0·051	Cords (1988)
Colobus angolensis	No	High	Low	—	—	—	0·218	−0·305	Groves (1973)
Colobus badius[c]	Yes	High	High	No	Low	High	0·226	−0·301	Struhsaker (1975); Isbell (1984)
Colobus guereza	No	High	Low	No	Low	High	0·370	0·046	Marler (1969, 1972); Oates (1977)
Colobus polykomos	No	High	High	No	Low	High	0·345	−0·170	Marler (1972)
Colobus satanus	No	High	High	—	—	—	0·052	−0·432	Struhsaker (1969); McKey & Waterman (1982)
Colobus verus	No	High	High	—	—	—	0·299	−0·358	Oates (1988)
Erythrocebus patas	No	High	Low	No	High	High	0·613	0·030	Chism & Rowell (1988)
Gorilla gorilla	No	High	Low	No	Low	High	−0·188	−0·565	Stewart & Harcourt (1987)
Hylobates concolor	No	High	Low	No	High	Low	0·481	0·322	Schilling (1984); Goustard (1984); Mitani (1985)
Hylobates hoolock	No	High	Low	No	High	Low	0·337	0·319	Roonwal & Mohnot (1977); Mitani (1985)
Hylobates klossi	No	High	Low	No	High	Low	0·280	0·324	Tilson & Tenaza (1982); Tenaza & Tilson (1985); Mitani (1985)
Hylobates lar	No	High	Low	No	High	Low	0·433	0·309	Mitani (1985); Palombit (1993)
Hylobates syndactylus	No	High	Low	No	High	Low	0·348	0·184	Chivers (1978); Curtin & Chivers (1978); Mitani (1985)
Macaca fascicularis	No	High	High	Yes	High	High	0·678	−0·031	Roonwal & Mohnot (1977); Wheatly (1982); Samuels *et al.* (1987); Walters & Seyfarth (1987)
Macaca fuscata	No	High	High	Yes	High	High	0·257	−0·372	Walters & Seyfarth (1987)

Table 1 ***Continued***

Species	Males			Females			Residuals[a]		References
	C[b]	I	F	C	I	F	Males	Females	
					Catarrhines				
Macaca mulatta	No	High	High	Yes	High	High	0·297	−0·348	Vessey & Meikle (1987); Ehardt & Bernstein (1986); Walters & Seyfarth (1987)
Macaca nemestrina	No	High	High	Yes	High	High	0·686	−0·076	Roonwal & Mohnot (1977); Caldecott (1986); Ehardt & Bernstein (1986); Walters & Seyfarth (1987)
Macaca nigra	No	High	High	—	—	—	0·715	−0·093	Shively *et al.* (1982); Caldecott (1986)
Macaca sinica	No	High	High	Yes	High	High	0·461	−0·002	Dittus (1986, 1988)
Mandrillus leucophaeus	No	High	High	—	—	—	1·028	−0·205	Stammbach (1987)
Miopithecus talapoin	No	High	High	—	—	—	0·431	−0·026	Wolfheim (1977); Gautier-Hion (1978); Keverne *et al.* (1978); Rowell (1988)
Nasalis larvatus	No	High	High	—	—	—	0·231	−0·320	Yeager (1990)
Pan paniscus	No	Low	High	No	Low	High	−0·384	−0·645	Nishida & Hiraiwa-Hasegawa (1987)
Pan troglodytes[d]	Yes	High	High	No	Low	High	−0·049	−0·306	Morin (1993)
Papio cynocephalus	No	High	High	Yes	High	High	0·394	−0·444	Smith & Whitten (1988); Shopland (1982); Melnick & Pearl (1987); Samuels *et al.* (1987); Walters & Seyfarth (1987)
Papio hamadryas	No	High	High	No	High	High	0·560	−0·283	Stammbach (1987)
Pongo pygmaeus	No	High	Low	No	Low	Low	−0·094	−0·350	Rodman & Mitani (1987)
Presbytis comata	No	High	Low	—	—	—	0·125	−0·281	Ruhiyat (1983)
Presbytis cristata	No	High	Low	No	Low	High	0·308	−0·201	Bernstein (1968); Wolfe & Fleagle (1977); Roonwal & Mohnot (1977)
Presbytis entellus	No	High	High	No	Low	High	0·264	−0·168	Blaffer Hrdy (1977, 1981); Newton (1986, 1987, 1988)
Presbytis melalophos	No	High	Low	No	Low	High	0·091	−0·427	Curtin (1976); Curtin & Chivers (1978); Johns (1985); Bennet (1986)
Presbytis obscura	No	High	Low	No	Low	High	0·152	−0·377	Curtin (1976); Curtin & Chivers (1978)
Presbytis pileata	No	High	Low	—	—	—	0·241	−0·314	Roonwal & Mohnot (1977); Green (1981); Islam & Husain (1982)
Presbytis potenziani	No	High	Low	No	Low	Low	0·088	−0·257	Tilson & Tenaza (1982); Tenaza & Tilson (1985)
Presbytis rubicunda	No	High	Low	—	—	—	0·035	−0·371	Jatna *et al.* (1986); Davies (1987)

Table 1 ***Continued***

Species	Males			Females			Residuals[a]		References
	C[b]	I	F	C	I	F	Males	Females	
					Catarrhines				
Presbytis vetulus	No	High	Low	—	—	—	0·221	−0·346	Rudran (1973); Roonwal & Mohnot (1977); Hladik (1978); Manley (1978)
Simias concolor	—	—	—	No	Low	High	—	−0·462	Watanabe (1981)
Theropithecus gelada	No	High	High	Yes	High	High	0·773	−0·261	Stammbach (1987)
					Platyrrhines				
Alouatta caraya	No	High	High	—	—	—	0·083	−0·167	Thorington *et al.* (1984)
Alouatta palliatta	No	High	High	No	High	High	0·061	−0·278	Milton (1978); Glander (1992)
Alouatta seniculus	No	High	High	No	Low	High	0·005	−0·323	Neville (1972); Klein & Klein (1975); Glander (1992)
Aotus trivirgatus	No	High	Low	No	High	Low	−0·135	−0·285	Wright (1985)
Ateles geoffroyi[d]	Yes	High	High	No	Low	High	−0·126	−0·570	Fedigan & Baxter (1984); Chapman (1990)
Ateles paniscus[d]	Yes	High	High	No	Low	High	−0·030	−0·436	McFarland Symington (1990); White (1986)
Brachyteles arachnoides[d]	Yes	High	High	No	Low	Low	−0·403	−0·646	Milton (1985); Strier (1986, 1992); Lemos de Sa & Glander (1993)
Callicebus moloch	No	Low	Low	No	Low	Low	−0·287	−0·266	Kinzey *et al.* (1977); Robinson *et al.* (1987)
Callicebus torquatus	No	Low	Low	No	Low	Low	−0·467	−0·510	Defler (1983); Robinson *et al.* (1987)
Cebuella pygmaea	No	Low	Low	No	Low	Low	−0·057	−0·107	Hernandez-Camacho & Cooper (1986); Goldizen (1987)
Cebus apella	No	High	High	Yes	High	High	0·398	0·119	Janson (1984)
Lagothrix lagotricha	No	Low	High	—	—	—	0·119	−0·399	Leo Luna (1980)
Saguinus fuscicollis	No	Low	High	No	High	Low	0·136	0·145	Goldizen & Terborgh (1986)
Saguinus midas	No	Low	High	No	High	Low	−0·035	0·015	Goldizen (1987)
Saguinus oedipus	No	Low	High	No	High	High	0·038	0·050	Dawson (1977); Neyman (1977)
Saimiri oerstedi	No	High	Low	No	Low	High	0·176	−0·284	Boinski (1987); Mitchell *et al.* (1991)
Saimiri sciureus	No	High	Low	Yes	High	High	0·155	−0·115	Klein & Klein (1975); Mitchell *et al.* (1991)
					Prosimians				
Arctocebus calabarensis	No	Low	Low	No	Low	Low	−0·217	−0·105	Charles-Dominique (1977)
Avahi laniger	No	Low	Low	No	Low	Low	−0·625	−0·649	Harcourt (1991)
Cheirogaleus medius	No	Low	Low	No	Low	Low	−0·143	−0·082	Petter (1988)

Table 1 ***Continued***

Species	Males			Females			Residuals[a]		References
	C[b]	I	F	C	I	F	Males	Females	
					Prosimians				
Eulemur coronatus	No	High	High	No	High	High	0·174	0·072	Kappeler (1992)
Eulemur fulvus	No	High	High	No	High	High	0·291	0·128	Pereira *et al.* (1990)
Eulemur macaco	No	High	High	No	High	High	0·175	0·140	Colquhoun (1993)
Eulemur mongoz	No	High	Low	No	Low	Low	0·201	0·028	Harrington (1978)
Eulemur rubriventer	No	High	Low	No	Low	Low	0·186	0·155	Overdorff (1991)
Galago elegantulus	No	Low	Low	No	Low	Low	0·117	0·063	Charles-Dominique (1977)
Galago moholi	No	Low	Low	No	Low	Low	−0·014	−0·023	Harcourt & Bearder (1989)
Galago senegalensis	No	Low	Low	No	Low	Low	0·031	−0·074	Bearder & Doyle (1974)
Galagoides alleni	No	Low	Low	No	Low	Low	−0·049	−0·218	Charles-Dominique (1977)
Galagoides demidovii	No	Low	Low	No	Low	Low	0·055	−0·135	Charles-Dominique (1977)
Galagoides zanzibaricus	No	Low	Low	No	Low	Low	0·022	−0·027	Harcourt & Nash (1986)
Hapalemur griseus	No	Low	Low	No	Low	Low	−0·268	−0·249	Petter & Peyrieras (1975)
Indri indri	No	Low	Low	No	Low	Low	−0·354	−0·453	Pollock (1979)
Lemur catta	No	High	High	No	High	High	0·161	−0·006	Kappeler (1990*b*)
Lepilemur mustelinus	No	Low	Low	No	Low	Low	0·199	−0·075	Russell (1977)
Lepilemur ruficaudatus	No	Low	Low	No	Low	Low	0·163	−0·036	Russell (1977)
Loris tardigradus	No	Low	Low	No	Low	Low	0·170	0·184	Petter & Hladik (1970)
Microcebus murinus	No	High	Low	No	Low	Low	−0·013	0·059	Peges-Feuillade (1988)
Microcebus rufus	No	High	Low	No	Low	Low	−0·188	−0·118	Harcourt & Thornback (1990)
Otolemur crassicaudatus	No	Low	Low	No	Low	Low	0·167	0·023	Bearder & Doyle (1974)
Otolemur garnettii	No	Low	Low	No	Low	Low	−0·078	−0·100	Nash & Harcourt (1986)
Perodicticus potto	No	Low	Low	No	Low	Low	0·089	0·065	Charles-Dominique (1977)
Propithecus diadema	No	Low	High	No	Low	High	−0·390	−0·278	Wright (1988)
Propithecus verreauxi	No	High	High	No	Low	High	−0·194	−0·259	Richard (1992)
Varecia variegata	No	High	Low	No	High	Low	0·263	0·222	Kaufman (1991)

[a]Residuals are from an isometric line passed through a data set treating male and female values as separate points.
[b]Abbreviations are as follows: C is coalitions, I is intensity, and F is frequency.
[c]Classification of coalitions in males is tentative, see text.
[d]Classification of coalitions in males is for between-group behavior. Within-group behavior is low-intensity, high-frequency.

It is now generally accepted that both male and female pithecine primates have canines that are specialized for opening hard objects (Plavcan & van Schaik, 1992; Greenfield, 1992*c*). For this reason, we exclude pithecines from the analysis.

Methods controlling for taxonomic artifacts

We first examined non-phylogenetically controlled species values. This corresponds to the hypothesis that canine size is under constant selective pressure in all species, and responds "instantaneously" to changes in selective pressure (Felsenstein, 1985). While this is probably an oversimplified view of evolution, we feel that examination of species residuals is important for preliminary hypothesis testing, and for understanding the total variation in canine size among species of various behavioral classifications. Species residuals were examined using standard ANOVA techniques in the SYSTAT package (Wilkinson, 1990).

Canine size is correlated with phylogeny in both sexes (Plavcan, 1990, 1993; Greenfield & Washburn, 1991). Numerous methods for controlling for taxonomic artifacts (or phylogenetic inertia) in comparative analyses have been and continue to be proposed and debated (see Harvey & Pagel, 1991, for a relatively recent summary). While there is no consensus as to which method works best, most agree that some effort must be taken to control for taxonomic artifacts. To control for taxonomic artifacts, we employed three alternate methods: a modification of the phylogenetic contrast method (Felsenstein, 1985; Pagel, 1992), repetition of the analysis within taxonomic groups, and analysis using means for species within genera that share the same behavioral classifications.

In principle, we preferred a modification of the phylogenetic contrast method (Felsenstein, 1985; Harvey & Pagel, 1991; Pagel, 1992) over other methods. This method is extensively discussed in Harvey & Pagel (1991) and Pagel (1991); the latter paper providing details of how to calculate contrasts. Ideally, the method compares values for a trait only between taxa that share a common ancestor, under the rational that differences between two such taxa must have evolved independently. Contrasts are calculated between terminal taxa in a phylogeny, and also between trait values that are reconstructed for nodes in the known phylogeny (Pagel, 1992). Unfortunately, the method requires not only a known phylogeny for a group, but also knowledge of the branch lengths within the phylogeny. While catarrhine phylogeny is relatively well known, platyrrhine and prosimian phylogeny remains uncertain. It follows that branch lengths for most taxa are highly uncertain at best, and that parametric statistical tests performed on contrasts must be approached with great caution.

Development of the phylogenetic contrast method has focused on comparisons involving continuous characters. Our data are both continuous and categorical. Ideally, contrasts for the continuous variable can be calculated and compared to the categorical data using standard ANOVA techniques. However, categorical data—especially the behavioral data—present considerable difficulties in reconstructing nodes for contrasts that undermines confidence in applying statistical procedures to the contrasts. Given two sister taxa that differ in a behavioral trait, there is no way to reconstruct the ancestral node without reference to an outgroup. However, in a number of cases in our data set there is no clear outgroup that can be used, since the outgroup is composed of taxa that are highly variable in the behavioral classification. In these cases, a hypothetical set of reconstructed ancestral character states can easily be generated under the assumption of parsimony, but we are highly wary of performing statistical tests on such hypothetical data. In our view, this lends an unjustifiable impression of statistical rigor to the analysis. No matter how the ancestral characters states are reconstructed, the less confidence there is in the accuracy of the reconstruction, the less confidence there should be

in validity of statistical evaluations of the contrasts. Another alternative—treating categorical variables as continuous character states to reconstruct nodes—creates meaningless contrasts of non-existent behavioral variables. The final alternative is commonality. When all or the overwhelming majority of species within a taxon share a character state, it can be assumed with reasonable confidence that the common ancestor also shared that state. Beyond obvious cases, though, commonality is a poor criterion for reconstructing ancestral character states, limiting the use of commonality for this purpose.

In spite of these problems, we still find value in carrying out phylogenetic contrasts. Comparisons between terminal taxa or clades of species sharing the same categorical classifications in a phylogeny present a series of independent tests of hypothesis, and especially allow a detailed examination of relative canine size among species that is not allowed by other methods.

We treated our canine data following the algorithm recommended by Pagel (1992), setting branch lengths equal to one (most branch lengths are unknown for our data set). Character states of the behavioral variables were reconstructed for each node using a commonality criterion; that is, an ancestral value was assigned when all species within a genus shared that character. Exceptions were made only when evidence was strong that a particular species (or node within a larger group) was derived in relation to the other species. To do otherwise would introduce a mixture of alternatives that would render the analysis meaningless. In practice, most contrasts are between congeneric species or species groups which are uniform in behavior, greatly simplifying the analysis and avoiding such problems. By eliminating contrasts between most higher nodes in the phylogenies (where ancestral character states are uncertain), we have sacrificed sample size to concentrate on the most conservative contrasts.

Where infrageneric relations are uncertain (as in most primate genera with more than two species), we represented species as a polychotomy. For analysis of contrasts, we resolve polychotomies in two ways (following the recommendations of Harvey & Pagel, 1991). First, commonly recognized species groups within genera were grouped together. Where taxonomic relations were too uncertain for this procedure, species sharing similar behavioral attributes were grouped together in order to create the minimum number of contrasts with the taxon. This rendered the analysis conservative in the number of contrasts.

Platyrrhine and prosimian phylogenetic relationships are highly uncertain. We therefore recalculated contrasts for a series of published phylogenies for platyrrhines (Rosenberger, 1979, 1981; Ford, 1986; Kay, 1990) and prosimians (Rumpler & Dutrillaux, 1986; Tattersall & Schwartz, 1991, 1992; Ishak *et al.*, 1992; Jung *et al.*, 1992; Macedonia & Stanger, 1992; Stanger & Macedonia, 1992). In practice, there were only minor differences in the contrasts among different phylogenies.

Because we have clear predictions about the relation between relative canine size and our behavioral classifications, contrasts were calculated between species or nodes on the basis of the predicted direction of change. Thus, the value of a low-intensity node was subtracted from that of a high-intensity node, since low-intensity species should have smaller canines than high-intensity species. A negative value thus contradicts the hypothesis, while a positive value supports it. To control for interactions, we calculated contrasts only between those species or nodes that differed in a single categorical variable. Contrasts between nodes that differed in two behavioral variables (in particular the frequency effect was ignored for a number of contrasts examining the effect of intensity, since we find little evidence of a frequency effect in females) are noted in the summary table.

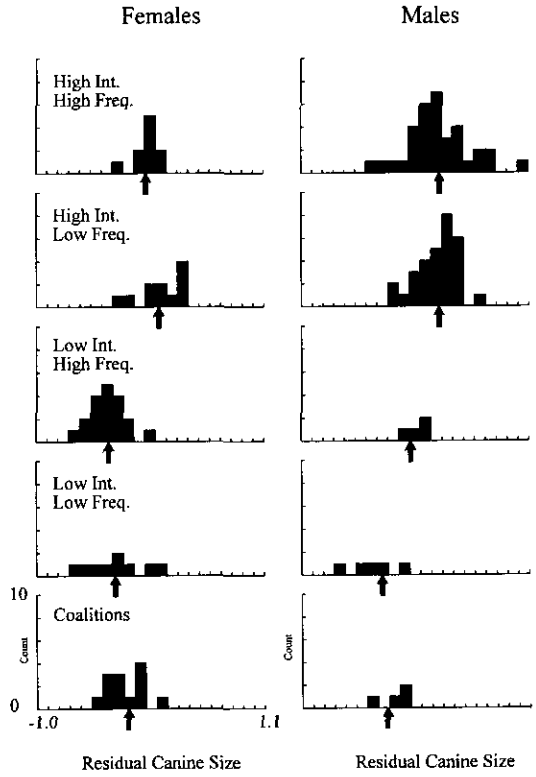


Figure 2. Frequency histograms of female and male relative maxillary canine crown height within behavioral classifications. Arrows indicate mean values for each group. Relative maxillary canine crown height is quantified by the residuals from an isometric line passed through the entire data set, treating males and females as separate points. The scale is the same for all histograms.

While the phylogenetic contrasts are highly informative for examining the details of the relation between canine size and the behavioral variables, the uncertain branch lengths and small sample sizes preclude meaningful statistical treatment. Therefore, we also controlled for phylogenetic effects using two simple alternative methods. First, all analyses were repeated in a series of lower taxonomic groups (Harvey & Pagel, 1991). This method controls for group differences in relative canine size (Plavcan, 1993), but has the disadvantage that sample sizes become quite small in lower taxonomic groups. Second, within each genus relative canine size was averaged for all species of the same behavioral classifications (Pagel & Harvey, 1988). This method preserves valuable categorical information at the lowest taxonomic level while reducing over-representation of speciose genera. The data so produced will be referred to in the text as the "taxonomically controlled data".

The three methods that we use to control for taxonomic artifacts compliment one another in how they control for the effects of phylogeny. Repetition of the analyses within taxonomic groups controls for consistent differences among higher taxa, analysis using means for species within genera controls for over-representation of particularly diverse genera, and phylogenetic contrasts allow a detailed evaluation of the hypotheses by only comparing canine size between sister taxa.

Table 2 **Analysis of species values**

Effect	*n*	F	*P*
Females:			
Coalitions	23	5·668	0·027
Intensity	64	31·825	<0·001
Frequency		11·364	0·001
Interaction		0·029	NS
Males:			
Coalitions	33	19·238	<0·001
Intensity	87	30·526	<0·001
Frequency		0·870	NS
Interaction		0·199	NS

Effect of coalitions was tested using one-way ANOVA of coalitionary species *vs.* high-intensity, high-frequency species. Effects of intensity and frequency were tested using two-way ANOVA of non-coalitionary species.

Dimorphism

To evaluate the association between competition in both sexes on canine dimorphism, we directly compared males and females using a matrix of behavioral classifications (25 cells in total, calculating dimorphism as the absolute difference between the sexes in canine size residulas from the pooled sample regression. This produced a series of size-corrected estimates of canine dimorphism. Cells across rows and columns were compared to summarize how variation in both male and female competition influence the general pattern of dimorphism. Species in which competition was classified in only one sex were not included in this analysis. Small sample sizes within the cells limits statistical treatment of the data, and so we can only interpret these comparisons with reference to the other analyses. Because behavioral or metric data were not available for both males and females, dimorphism could only be compared for 68 species.

Results

Competition

Species values. Females: Two-way ANOVA of female residual canine size for the intensity and frequency of conflicts and their interaction (using only non-coalitionary species) reveals a significant effect of intensity, and frequency (Figure 2 and Table 2). The frequency effect, however, is in the opposite direction to that predicted—low frequency females have on average larger canines than high frequency females. These results are repeated using the taxonomically controlled data (Table 3), and within all lower taxonomic levels except the ceboids, where the sample size is small.

Compared to other high-intensity, high-frequency species, female canines are significantly smaller in species that maintain dominance hierarchies through coalitions, as predicted (Figure 3, Table 2). Differences are not significant in either the taxonomically controlled data (Table 3), or within lower taxonomic levels, though differences are always in the predicted direction.

Males: Among males, the effects of intensity and coalitions are highly significant (Figure 3, Table 2). The effect of intensity is particularly strong, with minimal overlap in canine size between males of the two intensity classes. The intensity effect is significant within all lower taxonomic groups except ceboids. Using the taxonomically controlled data, the effects of intensity and coalitions are significant (Table 3). The frequency effect is not significant for all

Table 3 **Analysis of taxonomically controlled data**

Effect	*n*	F	*P*
Females:			
Coalitions	14	1·536	NS
Intensity	29	18·878	<0·001
Frequency		1·149	NS
Interaction		0·163	NS
Males:			
Coalitions	18	11·033	0·004
Intensity	34	19·137	<0·001
Frequency		4·251	0·048
Interaction		0·147	NS

Effect of coalitions was tested using one-way ANOVA of coalitionary species *vs.* high-intensity, high-frequency species. Effects of intensity and frequency were tested using two-way ANOVA of non-coalitionary species.

primates (Figure 3, Table 2) or the taxonomically controlled data (Table 3), but is significant within anthropoids, cercopithecoids, platyrrhines, and ceboids.

Phylogenetic contrasts. Females: The two valid contrasts for the effect of female coalitions are in the predicted direction (Table 4). One possible exception to the coalition hypothesis is *Cebus apella*, which has relatively large canines in comparison to most other platyrrhines. The highly uncertain phylogenetic position of this species precludes any meaningful contrast. Two factors may explain the anomalous canines of *Cebus*. First, they may not in practice be characterized by coalitionary competition, because they typically occur in very small groups (van Schaik, pers. obs.). Second, the large canines of *Cebus* could reflect a dietary specialization similar to that of the pitheciines. *Cebus* has been observed in the wild to open hard palm nuts, though further observations are necessary to discern whether this species actually uses the canines for this purpose.

We find no support from the phylogenetic contrasts for the frequency effect in females. Four of the eight contrasts are negative (Table 4).

All intensity contrasts for females are in the predicted direction (Table 3). This pattern holds regardless of frequency classification. Even species that appear anomalous using species values produce contrasts in the predicted direction. Thus, although *Aotus* has smaller canines than most other high-intensity species (and appears to be an outlier among the species values), they are substantially larger than those of closely related low-intensity species. Likewise, those prosimians classified as low-intensity have smaller canines than those classified as high-intensity (primarily the lemurs).

Males: For males, both coalition contrasts are in the predicted direction (Table 4). One coalitionary species (not listed in Table 4) is notable as a possible exception to the hypothesis, even though there are no valid contrasts. *Pan troglodytes* possesses canines relatively larger than either *Gorilla* or *Pongo* (Table 1), both of which are assessed as high-intensity, low-frequency species. However, *Pan paniscus* (assessed as low-intensity, high-frequency) possesses smaller canines than any other hominoid, making such a comparison difficult to evaluate.

All intensity contrasts for males are in the predicted direction (Table 4). This holds, even when the effect of frequency is ignored. Especially notable are the lemuriformes, in which *Lemur* and *Eulemur* show canines substantially larger than both *Lepilemur* (Tattersall & Schwartz, 1991, 1992), and *Hapalemur*, which is either the sister taxon of *Lemur* (Ishak *et al.*, 1992; Jung

Table 4 Phylogenetic contrasts

Contrast	Value	Comments
Females		
Coalitions:		
C. aethiops vs. other *Cercopithecus*	0·019	
Cercopithecus vs. Papionines	0·102	Assumes no coalitions in ancestral *Cercopithecus*
Frequency:		
Pongo vs. other apes	−0·110	
Presbytis potenziani vs. P. melalophos	−0·120	
Saguinus oedipus vs. S. fuscicollis	−0·066	
Brachyteles vs. Ateles	0·090	Rosenberger (1979) and Ford (1986)
Eulemur coronatus vs. E. rubriventer	−0·059	
E. mongoz vs. other *Eulemur*	0·068	Based on Stanger & Macedonia (1992) phylogeny
E. rubriventor/mongoz vs. E. macaco	0·050	Based on Tattersall & Schwartz (1991) phylogeny
Propithecus vs. Indri/Avahi	0·164	
Intensity:		
Alouatta palliatta vs. A. seniculus	0·032	
Aotus vs. Callicebus	0·066	Based on Rosenberger (1979) & Ford (1986) phylogenies
Lemurines *vs. Hapalemur*	0·176	Based on Stanger & Macedonia (1992) phylogeny
Lemur catta vs. Hapalemur	0·173	Based on Rumpler & Dutrillaux (1986) phylogeny
Lemurines *vs.* Cheirogaleids	0·116	Based on Rumpler & Dutrillaux (1986) phylogeny
Lemurs *vs. Lepilemur*	0·148	Based on Tattersall & Schwartz (1991) phylogeny ignores frequency (confounding)
Hylobates vs. great apes	0·369	Ignores frequency (conservative)
Cebuella vs. Saguinus	0·099	Ignores frequency (confounding)
Aotus vs. atelines	0·116	Ignores frequency (conservative), based on Kay phylogeny
Males		
Coalitions:		
C. badius vs. other *Colobus*	0·012	Assumes high frequency in ancestral *Colobus*
Brachyteles vs. Alouatta	0·274	Based on Kay (1990) phylogeny
Frequency:		
C. aethiops vs. other *Cercopithecus*	0·015	
papionines *vs.* cercopithecini	0·021	
Colobus angolensis/guereza vs. C. satanas/polykomos	−0·055	
Presbytis entellus vs. P. vetulus	0·031	
Nasalis vs. Presbytis	0·028	Assumes low frequency primitive for *Presbytis*
Cebuella vs. Saguinus	0·047	
Cebus apella vs. Saimiri	0·150	Based on Rosenberger (1979) phylogeny
Eulemur rubriventor vs. E. coronatus	−0·009	
Propithecus vs. Indri/Avahi	0·115	Based on Stanger & Macedonia (1992) phylogeny
Eulemur mongoz vs. E. coronatus/fulvus	0·021	Based on Stanger & Macedonia (1992) phylogeny
E. rubriventor/mongoz vs. E. macaco	−0·018	Based on Tattersall & Schwartz (1991) phylogeny
Intensity:		
Aotus vs. Callicebus	0·153	Based on Rosenberger (1979) & Ford (1986) phylogenies
Microcebus vs. Cheirogaleus	0·022	
Saimiri vs. callitrichines	0·130	Ignores frequency (conservative), based on Kay (1990) phylogeny
Cebini *vs.* callitrichines	0·191	Ignores frequency (conservative), based on Rosenberger (1979) phylogeny
Lemur catta vs. Hapalemur	0·305	Ignores frequency (confounding), based on Rumpler & Dutrillaux (1986) phylogeny
Lemurids *vs.* Indrids	0·291	Ignores frequency effects, assumes high intensity primitive for lemurids, based on Rumpler & Dutrillaux (1986) phylogeny
Propithecus verreauxi vs. P. diadema	0·195	
Lemur *vs. Lepilemur*	0·026	Based on Tattersall & Schwartz (1991) phylogeny, ignores frequency (confounding)

For some intensity contrasts, the frequency effect was ignored. Those where the effect of frequency is opposite that of intensity are marked "conservative", those where both frequency and intensity are predicted to have the same effect are marked "confounding".

Table 5 Qualitative summary of evidence from all four methods of analysis for the effects of coalitions, intensity and frequency on male and female primates

Effect	Analysis			
	Species values	Within-taxon analyses	Taxonomically controlled data	Phylogenetic contrasts
Coalitions				
Males	Moderate	Moderate	Strong	Moderate
Females	Strong	Weak (NS)	Weak (NS)	Moderate
Intensity				
Males	Strong	Strong	Strong	Strong
Females	Strong	Strong	Strong	Strong
Frequency				
Males	Weak (NS)	Moderate	Moderate	Moderate
Females	None	None	None	None

For analysis of species values and taxonomically controlled data, "strong" indicates highly significant results, "moderate" indicates moderately significant results, "weak (NS)" indicates results that are not statistically significant, but are in the predicted direction, and "none" indicates no support for the hypothesis. For the within-taxon analyses, "strong" indicates consistently strong results at all taxonomic levels, "moderate" indicates significant results within most, but not all taxonomic levels, "weak (NS)" indicates non-significant results in most taxonomic levels, but group differences always in the predicted direction, and "none" indicates no support for the hypothesis. For the phylogenetic contrasts, "strong" indicates that all contrasts are in the predicted direction, "moderate" indicates that all contrasts are in the predicted direction, but there are too few contrasts for confidence, and "none" indicates that the contrasts are divided between supporting and denying the hypothesis.

et al., 1992), or a *Lemur/Eulemur* clade (Macedonia & Stanger, 1992). Likewise *Propithecus verreauxi*, which appears to be an outlier in the comparison of species values (note the relatively small size of the canines compared to other high-intensity species in Table 1), has distinctly larger canines than the both *P. diadema* and the *Indri/Avahi* group. *Aotus*, which also has relatively small canines compared to most other high-intensity species (Table 1), is also notable in possessing canines relatively larger than *Callicebus* (though the phylogenetic position of *Aotus* is still uncertain).

Phylogenetic contrasts provide weak support for the male frequency effect. Eight of eleven contrasts are in the predicted direction (Table 4).

In summary, the results of the phylogenetic contrasts parallel those of the species values, and in the case of some outliers which appear to contradict the analysis of species values, such as *Propithecus verreauxi* and *Aotus*, actually provide additional support. All contrasts for the coalition and intensity effects are in the predicted direction, while at best the contrasts provide weak evidence for a frequency effect in males only. Once again, the small sample sizes and lack of branch lengths for the contrasts precludes meaningful statistical tests, and this must be considered when assessing these results.

Table 5 qualitatively summarizes the results of the four methods used to evaluate the effects of coalitions, frequency and intensity on relative canine size in males and females. In both males and females, all analyses provide strong support for the intensity effect on relative canine size in both sexes. There is no support for a frequency effect in females from any analysis—in fact the effect is reversed from that predicted (see discussion). On the other hand, while there is only weak, non-significant support for the frequency effect in males when species values are analysed, the methods controlling for the effect of phylogeny all provide moderate support for the hypothesis. Finally the coalitions hypothesis receives a variable degree of support among the methods for males and females. Importantly, while there is variation in the strength of

support for the hypotheses among the various methods, the pattern of results is consistent no matter which method is used.

Dimorphism: Figure 3 provides a comparison of canine dimorphism for species within each of the 25 possible combinations of male and female behavioural classifications. Eleven of these cells are empty. The others provide a rough overview of how canine dimorphism changes with changes in male and female behavior. Effects of behavioral differences within each sex on sexual dimorphism can be compared across rows or columns comparing cells that differ in only one behavioral classification within one sex (for example, comparing cells that differ only in the classification of female intensity). Because of the complexity of interactions among the behavioral variables, there are almost no meaningful phylogenetic contrasts. Therefore we concentrate on the species values. Because of the complexity of the comparisons from Figure 3, a summary of the comparisons generated from Figure 3 is presented in Table 6.

If the classifications of contest features reflect similar selection pressures in both males and females, then species in the diagonal element of Figure 3 (cells A, G, M, S and Y) should be monomorphic. In all of them canine dimorphism is much smaller than in most other cells. The results are not an artifact of phylogeny—the cells contain mixtures of galagines, callitrichids, hylobatids and lemurids. Nevertheless, it is important to note that in all of these cells canine dimorphism persists, even though it is comparatively slight.

We predict that where male competition intensity is high canine dimorphism should be larger, and where female competition intensity is high, canine dimorphism should be smaller. For males, each comparison between cells for the effect of intensity follows this prediction (cells I *vs.* G, O *vs.* M and Y *vs.* W). For females, three of the four comparisons between cells are in the predicted direction (cells W *vs.* M, Y *vs.* O and S *vs.* I). We note, however, that differences in the intensity of male competition are associated with greater differences in canine dimorphism than for females.

We predict that where male competition frequency is high, dimorphism should be larger, and where female competition frequency is high, canine dimorphism should be smaller. Comparisons parallel the results of the phylogenetic contrasts and the species values. Where male frequency is higher, dimorphism is larger in two of three comparisons (cells D *vs.* E, N *vs.* O). Where female frequency is high, canine dimorphism is actually larger in each of four comparisons (cells F *vs.* K, I *vs.* N, R *vs.*W and S *vs.* X).

We predict that where male competition is coalitionary, dimorphism should be smaller, and where female competition is coalitionary, dimorphism should be larger. Two of three comparisons show this to be true (Males: cells K *vs.* O. Females: cells E *vs.* Y). One comparison shows that where female coalitions are present, dimorphism is smaller (cell D *vs.* X), but the cell with female coalitions contains only one species.

Overall, these comparisons appear to corroborate the hypothesis that variation in the intensity of competition in both sexes seems to have an obvious, predictable effect on sexual dimorphism. The context (coalitionary/non-coalitionary) of competition in both males and females also appears to affect the expression of dimorphism. On the other hand, the effect of frequency on dimorphism follows the predicted pattern only for males, and then in only two of three comparisons. While we could not generate enough meaningful phylogenetic contrasts to assure that these results are not confounded by taxonomic artifacts, the results nevertheless parallel those for the independent analyses of species values and phylogenetic contrasts in males and females remarkably well.

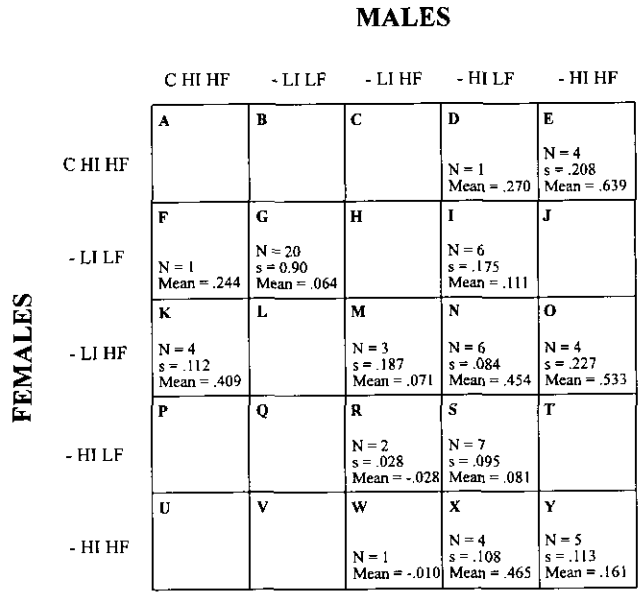


Figure 3. Matrix of male and female behavioral classifications, showing the number of species present in each cell, and the mean and standard deviation of canine dimorphism of all species in each cell. Dimorphism is calculated as the absolute difference in residual canine size (measured from an isometric line passed through a data set treating males and females as separate points) between males and females. Cells are labeled A-Y for ease of reference in the text. Abbreviations are as follows: (C) coalitions present; (-) coalitions absent; (HI) high intensity; (LI) low intensity; (HF) high frequency; (LF) low frequency.

Table 6 **Summary comparisons of sexual dimorphism in maxillary canine crown height between cells in Figure 3**

	Coalitions		Intensity		Frequency	
Males	0·124	(O-K)	0·047	(I-G)	0·369	(E-D)
			0·462	(O-M)	0·079	(O-N)
			0·171	(Y-W)	− 0·304	(Y-X)
Females	0·195	(X-D)	− 0·030	(S-I)	0·343	(N-I)
	− 0·478	(Y-E)	− 0·081	(W-M)	0·018	(W-R)
			0·011	(X-N)	0·384	(X-S)
			− 0·372	(Y-O)	0·165	(K-F)

Other factors

Our data provide some support for the hypothesis that large male canines are selected for predation defense, especially in savanna-dwelling species (Harvey *et al.*, 1978*b*; Rowell & Chism, 1986; Plavcan & van Schaik, 1992). Overall, substrate is associated with relative male canine size (n=94, F=7·612, P=0·001) but this result appears to be a taxonomic artifact reflecting general differences in relative canine size among higher taxa. Substantial variation in substrate occurs only with catarrhines, wherein the relation is not significant. Notably, however, the few savanna-dwelling species (*Papio*, *Erythrocebus* and *Theropithecus*) show larger canines than all other species in similar competition categories, providing modest support for the hypothesis.

Our data do not support several alternative explanations for variation in female canine size. As for males, relative female canine size might be larger in savanna-dwelling species due to increased predation risk (Harvey *et al.*, 1978*b*; Plavcan & van Schaik, 1992). This does not hold for females (n=94, F=0·726, P=0·396), possibly because only males actively defend groups against terrestrial predators. Two other alternatives make conflicting predictions about female canine size. Females living in multimale-multifemale groups may face greater competition over resources than those living in single-male groups. If so, they should have larger canines (Harvey *et al.*, 1978*a*). Alternatively, females living in single-male groups may need larger canines than those living in multimale groups because their offspring face a greater risk of infanticide (Haufstater & Hrdy, 1984). We found no difference in relative female canine size between single-male and multi-male species within the (female) high-frequency class (n=69, F=0·264, P=0·609). Finally, it may be argued that relative female canine size is larger in those prosimians showing female dominance over males. However, a recent comparative analysis (Kappeler, 1992) found no support for this idea.

Greenfield (1992*a*,*b*) suggests that interspecific variation in female canine size represents a correlated response to variation in male canine size (Lande, 1980). The correlation between residual male and female canine size is 0·51, explaining 26% of the variation in female canine size, consistent with the hypothesis that correlated response partly explains variation in female canine size.

We find little evidence of a general relation between canine size and diet among primates, lending little support to the hypothesis that a folivorous diet is associated with low levels of dimorphism (Milton, 1985). On the other hand, there is strong evidence that the canines of pitheciine primates are specialized for opening hard objects (Greenfield, 1992*c*). As mentioned above, our data also suggest that *Cebus* canines may be similarly influenced by diet.

Discussion

Competition and relative canine size

That the development of weaponry is correlated with the intensity of intrasexual contest competition in *both* sexes undermines models which either implicitly or explicitly view dimorphism exclusively as a product of sexual selection via male–male contest competition for access to mates. While our analysis clearly supports this hypothesis for male canine size, it also suggests that sexual selection alone does not explain the evolution of canine dimorphism in primates. Relative canine size among females is almost as variable as that among males (standard deviations of residual canine size are 0·23 and 0·30, respectively), so the forces that shape female canine size must play a substantial role in the evolution of canine dimorphism in primates. Our results suggest that canine dimorphism reflects, at least in part, a balance between selection for the development of weaponry in both sexes.

We find little evidence that our results represent a taxonomic artefact. The different ways of controlling for taxonomic artefacts produce a consistent pattern of results (Table 6). Support for the intensity effect in both males and females is consistently strong no matter how the data are treated or analysed. Evidence for the frequency effect is consistently weak to moderate for males, and lacking for females. Support for the coalitions effect varies from weak to strong, depending on the method of analysis, but in all cases the pattern of results is the same. The variation in the strength of support for the coalitions effect seems to reflect variation in sample size, with weak support indicating a lack of power in the statistical tests, rather than a

breakdown in the pattern of results. Our representation of species of primates is extensive, covering all genera except *Allenopithecus, Allocebus, Callimico, Daubentonia, Leontopithecus, Mirza, Phaner, Pygathrix, Rhinopithecus* and *Tarsius*. Thus, it is unlikely that additional data will alter the results of the analyses. On the other hand, our results should not be interpreted as suggesting that there is no correlation between relative canine size and phylogeny in primates. To the contrary, canine size is correlated with phylogeny in both sexes (Greenfield & Washburn, 1991; Plavcan 1990, 1993). Rather, our data suggest that *both* intrasexual competition and phylogeny are associated with variation in relative canine size in both sexes.

Our results strongly support the hypothesis that canine size in both sexes is associated with the intensity of intrasexual competition. No matter how the data are treated, species ranked as high-intensity consistently possess relatively taller canines than species ranked as low intensity.

Our analysis suggests that there is an effect of potential contest frequency on canine size in males. We previously hypothesized that a frequency effect may reflect selection for weaponry as display in males (Plavcan & van Schaik, 1992). In the high-frequency category males cannot gain exclusive access to a group of females, yet still compete intensely for dominance. For males to escalate every fight may be disadvantageous in this situation, since the winner may not be able to exclude the loser from the group, while a third male may actually benefit more than either contestant. This should favor a mechanism to determine the outcome of contests through display of weaponry, rather than escalation of every conflict (Packer & Pussey, 1985).

On the other hand, the effect of frequency on female canines is opposite to that predicted, with low-frequency females actually tending to have relatively larger canines than high-frequency females. That the mechanism described for males should not work for females makes sense. Females may compete intensely for access to resources, but where they are gregarious they usually do not attempt to evict others from their groups, because they actually benefit from group living. This is at least partly consistent with the observed reversal of the frequency effect in females. Solitary females that defend resources should face a higher fitness consequence of winning and losing fights than group-living females, which either do not compete intensely (as in many colobines), or compete in alliances.

Our analysis suggests that the highly exaggerated canine dimorphism seen in many anthropoid primates is a function not just of intense intermale competition for access to mates, but also of coalitionary competition among females. Those species in which both males and females show intense intrasexual competition, but only females regularly do so in large coalitions, show highly exaggerated canine dimorphism when compared to species in which females compete intensely, but not in coalitions. Among anthropoids, females commonly form matrilineal dominance hierarchies (indicating that coalitionary competition is important) in macaques, baboons, vervet monkeys (*Cercopithecus aethiops*), one species of squirrel monkey (*Saimiri sciureus*), capuchin monkeys (*Cebus apella*), and possibly mangabeys. Though competition among females in all of these species can be intense, with the exception of *Cebus* (which we suggest may have large canines in males and females in association with hard-object feeding), their canines tend to be relatively smaller than in species which compete intensely, but not in coalitions.

The effect of coalitions also may help explain why males of male-bonded species such as *Brachyteles, Pan troglodytes* and *Ateles* possess relatively small canines. It should be kept in mind, however, that coalitionary competition among the males of these species should occur between groups, rather than within, and that the evidence for intergroup competition is tenuous,

especially for the atelines (Strier, pers. comm.). Considering only within group behavior, these species would be classified as low-intensity, high-frequency. That their relative canine size is consistent with either classification leaves open the question of how important coalitions are in the evolution of male canine size.

It is interesting to note that, holding competition constant, males on average have relatively larger canines than females, as reflected by the persistent, slight dimorphism in those species in which males and females are classified in the same competition cells (Figure 3). This may reflect a number of factors. First, male canine size generally may be larger as a result of non-selected developmental effects (such as pleiotropic or hormonal effects). This would help explain why male canines tend to be slightly larger than those of females in the majority of species in which neither sex engages in high-intensity competition. Perhaps more importantly, though, our simple behavior dichotomies cannot distinguish differences in the strength of selection for weaponry between males and females. It is well known that males usually compete for access to mates, while females usually compete for access to resources such as food (Trivers, 1972; Wrangham, 1980). Thus, in general the fitness consequences of losing fights should be greater for males than for females, and selection for the development of weaponry should be stronger in males. This raises the important point that our classifications of competition are intended only to reflect the proximate behaviors that should be associated with selection (either natural or sexual) for the development of weaponry. The nature of competition among individuals varies widely among species, and a better understanding of the ultimate causes of selection for the development of weaponry, especially among females, should relate competition directly to patterns of resource utilization. Such a detailed examination is beyond the scope of this paper.

As previously mentioned, our intensity classification roughly parallels the distinction between scramble and contest competition (especially as concerns females). Because these two behavioral classifications are closely related to diet, it may seem paradoxical that we find no strong relation between canine size and diet, especially in females. The explanation lies in the imperfect relation between diet and competition intensity. For example, female orang-utans, though basically solitary, are relatively amicable towards one another, while female lar gibbons are not. Both, however, are primarily frugivorous. Likewise, species may compete over resources such as fruit, but still incorporate substantial amounts of leaves in their diets. Thus, while resource competition, and therefore diet, may ultimately influence competition and selection for the development of weaponry, in fact there is no consistent, one-to-one correspondence between diet and female canine size.

Our results do not exclude other factors as contributing to the evolution of dimorphism. For example, relative canine size in both sexes is correlated with phylogeny, though the exact mechanism producing phylogenetic correlations is unclear (Greenfield & Washburn, 1991; Plavcan, 1993). There is evidence that canines may serve as defensive weapons from predators in savanna-dwelling species (Harvey *et al.*, 1978*b*; Plavcan & van Schaik, 1992), and indeed our data show that males of savanna-dwelling species have larger canines than most of their close relatives regardless of competition type. Diet has clearly played a role in the evolution of pitheciine canine size and shape (Greenfield, 1992*c*), and cannot be excluded as a factor in other species (for example *Cebus*), though we have found no general relation between diet and either canine size (Plavcan, 1993, this study) or canine dimorphism (Plavcan & van Schaik, 1992). None of these factors is mutually exclusive, and a complete model should consider all factors together, although the nature and extent of trade-offs remain largely unknown.

Another factor that may help explain the evolution of canine dimorphism is correlated response (Lande, 1980; Greenfield, 1992*a,b*). Lande (1980) points out that if a character is autosomal and polygenic, then selection for change in one sex should produce a correlated change in the other, since genetic control of the expression of the character should be the same in both sexes. Only after an extended period of time can dimorphism evolve by a dissociation between the sexes of genetic control of the character. The finding of a correlation across species between male and female residual canine size partly supports this hypothesis (Greenfield, 1992*a,b*). However, canine tooth size dimorphism is an ancient trait, having been observed in Eocene notharctines (Krishtalka *et al.*, 1990) and Oligocene propliopithecids (Fleagle *et al.*, 1980), suggesting that there has been ample time under the Lande model (1980) for the effect of correlated response to abate. This, coupled with the fact that canine dimorphism is highly variable among primates suggests that correlated response plays only a minor secondary role in explaining canine dimorphism and especially female canine size in extant species (Greenfield, 1992*a,b*).

On the other hand, the correlation between male and female relative canine size may help explain why taxonomic correlations in relative canine size in either sex are not strongly repeated when canine dimorphism is examined, and suggests that estimates of canine dimorphism which compare males and females relative to each other actually control for the effects of phylogeny (Harvey *et al.*, 1978*b*). Thus, studies find only weak evidence of phylogenetic inertia in canine dimorphism, but strong phylogenetic correlations in relative canine size in both sexes (Kay *et al.*, 1988; Greenfield & Washburn, 1991; Plavcan & van Schaik, 1992; Plavcan, 1993).

Prosimian canine dimorphism

Prosimians are generally monomorphic in body size (Kappeler, 1990*a*, 1992) and canine size, yet they show a variety of mating systems. This presents a potentially disturbing contradiction to the sexual selection hypothesis, and demands explanation. Prosimian behavior is relatively poorly known, and many of our classifications are tentative pending thorough field studies. Nevertheless, according to our classification system, most male and female prosimians compete in roughly similar ways, and therefore should be characterized by low levels of dimorphism. The observation that, within each sex, competition intensity is associated with variation in relative canine size strongly suggests that canine size in prosimians is under similar selective pressures as in anthropoids. Therefore, these results suggest that prosimian canine monomorphism is consistent with the anthropoid pattern, in spite of the fact that variation in prosimian mating systems would suggest otherwise (cf. Kappeler, 1993).

A more intriguing observation is that, for high-intensity competition species, prosimian male canines are, on average, relatively smaller than those of anthropoids. On the one hand, this may represent a phylogenetic effect on canine size independent of any adaptive factor (Plavcan, 1993). On the other hand, van Schaik & Kappeler (1993) recently suggested that in the cathemeral gregarious prosimians (most of the high-intensity species in our sample), the pair bond forms the fundamental social unit, and that males of most lemuriformes are limited in the ability to gain exclusive access to multiple females as a result of intense estrous synchrony (the largely diurnal *Lemur catta*, as expected, is an exception: Pereira, 1991). If true, then prosimian males cannot gain the large reproductive payoffs from intrasexual competition that male anthropoids achieve, and the fundamental difference between anthropoids and prosimians in the development of canine dimorphism is the

relative strength of selection resulting from male–male competition. Along the same lines, Richard (1992) observes that, while male *Propithecus verreauxi* fight intensely during the mating season, female choice may play a substantial role in determining mating success for males. Thus, fighting success does not translate into reproductive success, and selection for weaponry should be reduced. This might explain why *P. verreauxi* does not develop canines as large as other high-intensity prosimians, even though they are substantially larger than their closest low-intensity relatives.

Mating system and sexual dimorphism

Our analysis suggests that mating system (as estimated by group composition) is an inadequate measure of the strength of intrasexual competition, and hence of sexual selection. Among primates, mating system is correlated with canine dimorphism only in that the males and females of most monogamous species tend to compete in broadly similar ways, and consequently show little dimorphism (Phillips-Conroy & Jolly, 1981; Rowell & Chism, 1986; Plavcan, 1990; Plavcan & van Schaik, 1992; Richard, 1992; Kappeler, 1992). Variation in the types and degrees of intrasexual competition is especially large among multimale-multifemale species. Such species include baboons, in which males compete intensely, while females compete within matrilineal dominance hierarchies, squirrel monkeys in which male–male competition is seasonal, while females either form matrilineal dominance hierarchies (*Saimiri sciureus*) or do not (*S. oerstedi*), gelada baboons, in which a multilevel society composed of single-male harems is formed, and *Brachyteles* which forms "fission-fusion" groups. In fact, this analysis suggests that the use of mating system to estimate the strength of sexual selection can actually confuse models of dimorphism, since it effectively ignores both the potential contribution of females to variation in dimorphism, as well as variation in male-male relations. However, we should also stress that the estimates of competition employed here, while an improvement over mating system, are still only crude generalizations of behavior and cannot capture the large and continuous variation in the actual frequency and intensity of intrasexual competition. Furthermore, this analysis should in no way be interpreted as indicating that mating system is somehow an invalid or useless concept. To the contrary, current models of mating systems are considerably more sophisticated than the broad classifications traditionally used to test the sexual selection hypothesis, and reflect a substantial increase in our understanding of mammalian behavior (Clutton-Brock, 1989; Davies, 1991).

Our analysis also suggests that studies of sexual dimorphism in any character should not assume that females always serve as a baseline of comparison for males. Our analysis strongly suggests that dimorphism varies as a consequence of factors that change canine size in both sexes. Analyses that seek to explain canine dimorphism as a function of factors that alter males relative to females cannot, as a consequence, fully explain the evolution of canine dimorphism. Logically, this should apply not only to canine tooth size dimorphism, but to other characters as well.

In this light, it is tempting to assert that ratio estimates of sexual dimorphism are fundamentally misleading for testing hypotheses of the causes of sexual dimorphism in any character. However, we do not believe that this is the case. The results of our analysis, after all, do not conflict with previous analyses that employed ratio estimates of dimorphism (Kay *et al.*, 1988; Plavcan & van Schaik, 1992). Furthermore, for characters such as body size it is difficult to derive independent, size-corrected estimates of size dimorphism for males and females (since size-corrected estimates are usually derived by comparison to body weight). Rather, a better

understanding of dimorphism is provided by considering both ratios and estimates of character variation in each sex separately whenever possible.

Canine reduction

Our data suggest that in the absence of selection for large canines, some mechanism or mechanisms act to reduce canine size. For example, the males of *Callicebus*, *Homo*, *Presbytis potenziani*, and *Brachyteles arachnoides* all possess small canines relative to the males of other closely related species. This suggests that the relatively small male canines of these species are derived, and that canine reduction occurs relatively rapidly in the absence of selection to maintain large canines. (Were it not to occur rapidly, we should expect to see some case or cases where males that do not compete intensely have canines as large as closely related species that do compete intensely.) Therefore, to some degree canine size must reflect a balance between selection to develop large canines, and selection to reduce the canines. However, it is important to emphasize that the mechanisms of canine reduction are not addressed in our model. As for most other studies of canine dimorphism, our model focuses on the selective mechanisms favoring the development and/or maintenance of large canines. We reiterate that the effect of coalitions is not an active mechanism for canine reduction. Rather, it helps explain why males or females of many species possess relatively small canines even though they fight with one another. A number of studies have suggested mechanisms for canine reduction (see Greenfield, 1992*b* for an extensive review), though none have gained wide acceptance as yet. At this point we can only infer that if there is selective pressure to reduce the canines in the absence of selection to maintain large canines, then, logically, large canines must be in some way costly to an individual. Several costs can be envisioned. Large canines may be metabolically expensive to develop, or they may interfere in some way with either food gathering or processing. Given the relatively small size of canines, we regard the former as unlikely, while we are aware of no support from either *in vitro* feeding studies or from field observations for the latter. Alternatively, large canines, to be functional, might require modifications to facial morphology that compromise the efficiency of the masticatory apparatus. The elucidation of the factors causing canine reduction remains a major task in studies of canine dimorphism, and is beyond the scope of this paper.

Studies of a variety of mammals have recognized that sexual dimorphism in mammals cannot be explained without considering female variation (i.e. Ralls, 1976, 1977; Shea, 1986; Brooks, 1991; Richard, 1992; Wright, 1993). Because canines in primates are easy to measure and clearly function as weapons, the analysis of canine dimorphism in primates provides a relatively simple example of how dimorphism is a product of evolution in both males and females. Our analysis suggests that similar principles govern the development of weaponry in both sexes, and there is no reason to suspect that this should not be the case in other taxonomic groups and other characters.

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